

ATLANTIC COMPUTATIONAL EXCELLENCE NETWORK

ACEnet: Free Cluster Computing For Researchers

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What is ACEnet?

- “Atlantic Computational Excellence Network”
- Shared regional resource for ...
 - ... high-performance computing
 - ... remote collaboration
 - ... visualization
- Hardware, software, and personnel

Remote Collaboration



Internet videoconferencing - inSORS & Access Grid

Dalhousie - Memorial - Mount Allison - Saint Mary's - Saint Francis Xavier
Acadia - New Brunswick + **other sites across Canada & the world**

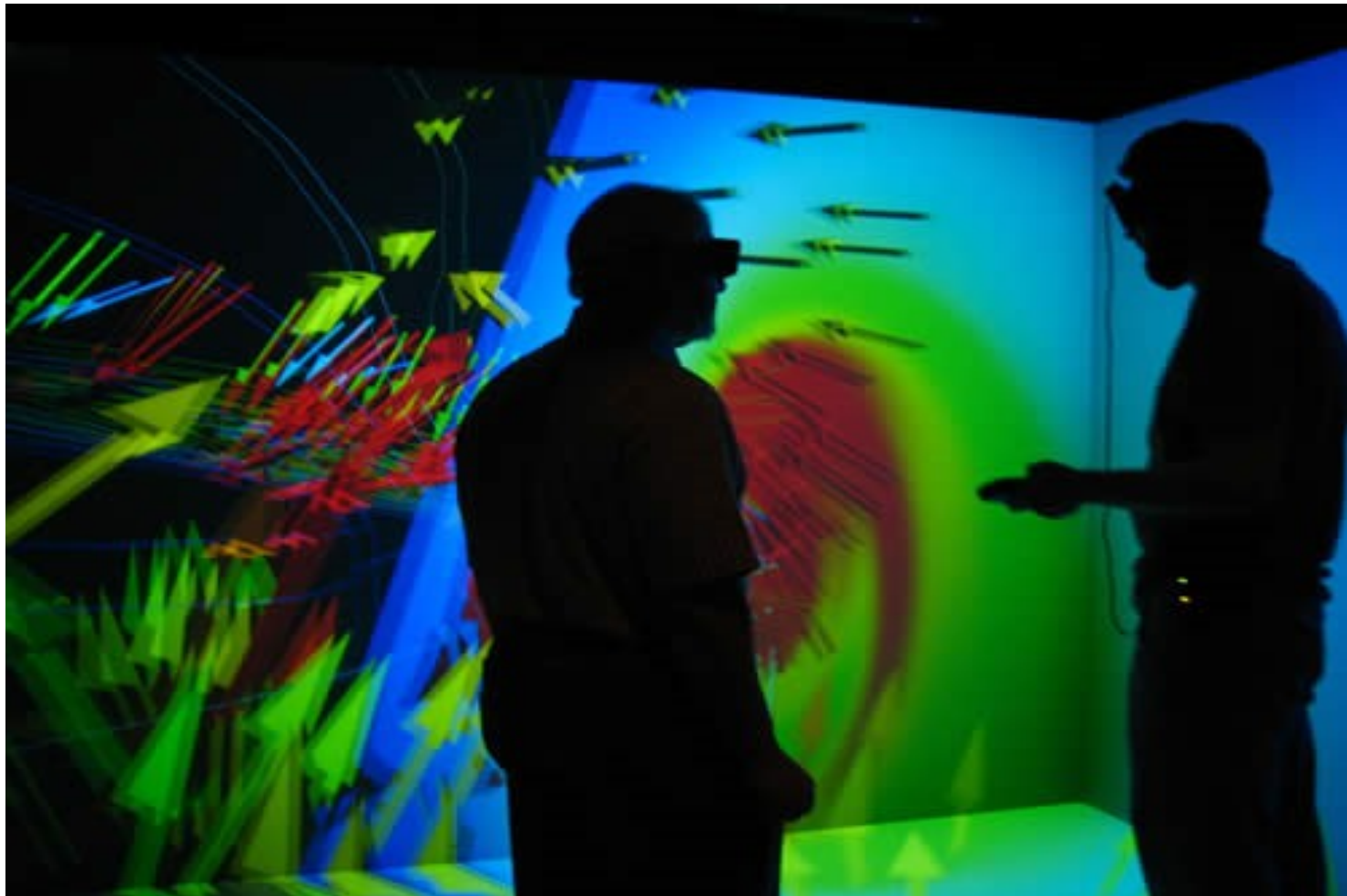


Advanced Visualization



Data cave

3D immersive data exploration facility @ Saint Mary's

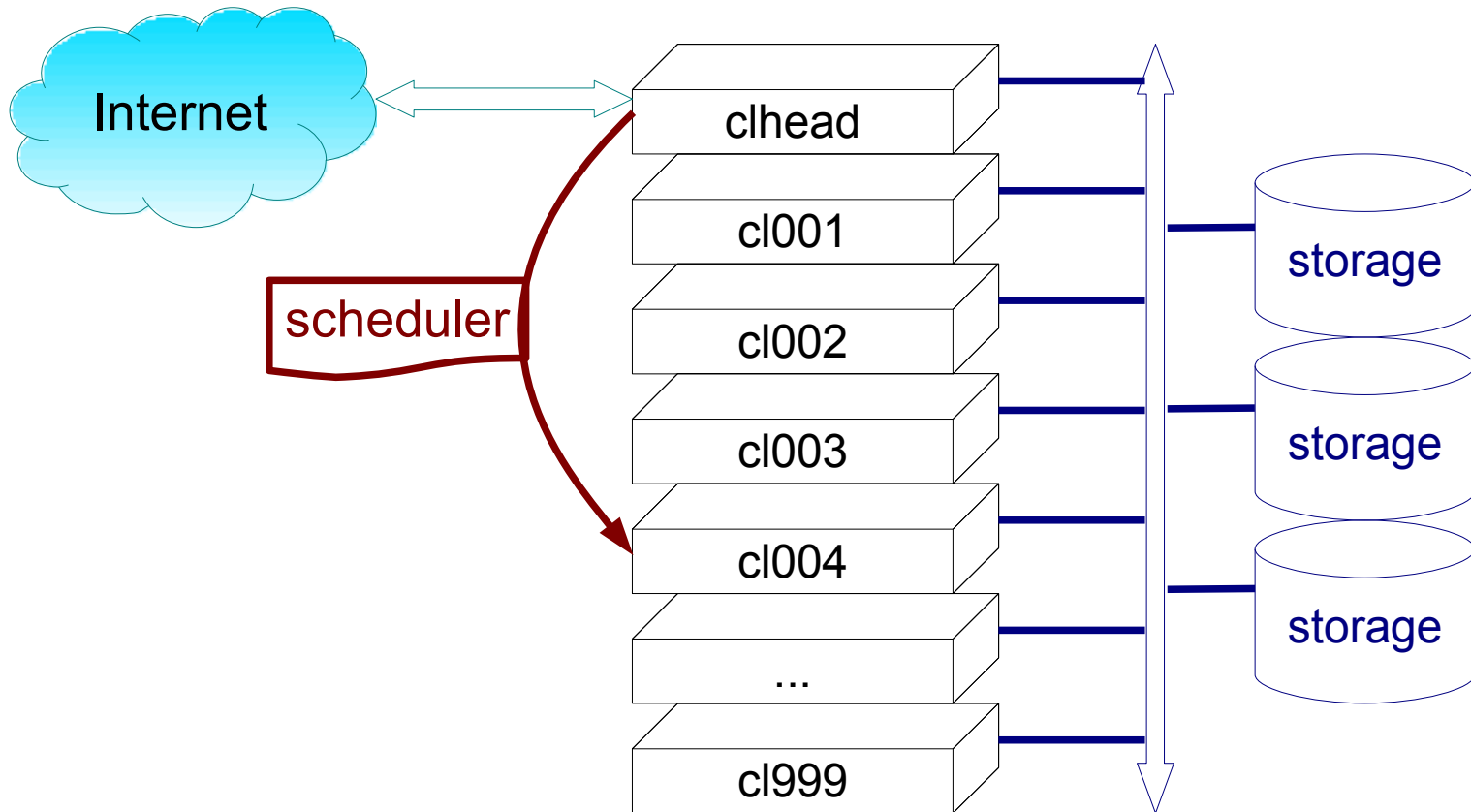


What's “High Performance Computing”?

- Many CPUs working on one problem
 - Many “serial” jobs running at once, or
 - many CPUs working in close coordination.
 - Communication between CPUs is key!
- Advantages
 - Faster results
 - Larger problems or more detailed simulations
- Principal tool: Computer **cluster**
 - Racks of regular computers, networked



Cluster workings



CPU and Memory Configurations

ACEnet clusters



Cluster	Nodes	Cores/node	RAM/node	Total cores
Brasdor	269	4	8G	1152
Courtenay	13	4	16G	52
Fundy	38	16	64G, 128G	636
	7	4	16G	
Glooscap	55	4	4G	844
	39	16	32G	
Mahone	138	4	16G, 64G	552
Placentia	154	16	32G, 64G	2884
	17	12	24G	
	54	4	16G, 64G	
	117	owned by individual groups		908

Storage Configurations

ACEnet clusters

Cluster	/home user quota	/globalscratch user quota	/nqs size	/scratch/tmp
Brasdor	47 Gb	238 Gb	14 Tb	20 - 330 Gb
Courtenay	--	--	2.0 Tb	100 Gb
Fundy	47 Gb	238 Gb	14 Tb	35 - 100 Gb
Glooscap	13 Gb	48 Gb	2.8 Tb	0 - 15 Gb
Mahone	13 Gb	238 Gb	14 Tb	50 - 100 Gb
Placentia	13 Gb	238 Gb	14 Tb	0 - 140 Gb
Cluster-local				Node-local

Applications

<http://www.ace-net.ca/wiki/Software>

<u>Molecular</u>	<u>Math&CS</u>	<u>Engineering</u>	<u>Earth &Ocean</u>	<u>Bio</u>
Gaussian	spin	ANSYS		mpiBlast
Gromacs	GAP	Fluent/	CDO	MrBayes
CPMD	DiVinE	Gambit	GMT	IM
Abinit	Macaulay-2	CFX	MM5	Migrate-n
NAMD	...	LS-DYNA		PhyML
WebMO		elmer-fem	<u>Astro</u>	PLINK
VMD			StarLab	
...			IRAF	

Software requests to:

support@ace-net.ca

Software Development

<http://www.ace-net.ca/wiki/Software>

<u>Languages</u>	<u>Suites</u>	<u>Parallel</u> <u>APIs</u>	<u>Libraries</u>	<u>Tools</u>
C/C++	PGI		LAPACK	TotalView
Fortran	Intel	Open MPI	FFTW	dbx, gdb
Perl	Gnu	OpenMP	GSL	Valgrind
Python	SunStudio	...	ACML	make
Java			netCDF	cvs, svn
...			HDF	git
			Boost	...
			...	

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Data Analysis



<http://www.ace-net.ca/wiki/Software>

R

Rmpi

Octave

Matlab*

*Distributed Computing Server

Python

Numpy

SciPy

Sage

pyMPI

pylab

matplotlib

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Operations Research?



<i>Package</i>	<i>Got it already?</i>	<i>Free $\approx$ Easy</i>	<i>Costs</i>
Python	Got it		
Octave	Got it		
MATLAB	Sort of...		
GLPK		Yes	
CPLEX			\$
ARENA			\$
Stella & iThink			\$

What's the cost?

- No money. It's paid for.
- Several \$million from
 - Canada Foundation for Innovation (CFI)
 - Atlantic Canada Opportunities Agency (ACOA)
 - Provinces (NS, Nfld, NB)
 - “in kind” contributions from Sun Microsystems*, and hosting universities

* Now Oracle

No really: What's it cost?

- Keep publishing.
 - and tell us about it, once a year
in conjunction with account renewal
- Help us out.
 - Local User Groups ← **user direction!**
 - Research Directorate, other committees
- Share the resource.
 - about 300 projects, 1000 users
 - jobs must use dynamic resource manager,
"Sun Grid Engine"

Training Available

Seminars on demand (~ 1 hr each)

- Introduction to ACEnet
 - Linux command line
 - Using Sun Grid Engine
 - Perl Programming
 - ...many more...
-
- Cross-Canada courses *via* videoconference
 - e.g. Parallel Programming course,
from HPCVL in Kingston, Ont., Feb 2011
 - Custom tutorials
 - ...*your interest here...*

Command-Line Interface



```
glooscap-head.cs.dal.ca - PuTTY
Using username "rdickson".
rdickson@glooscap.ace-net.ca's password:
Last login: Wed Mar 16 10:03:16 2011 from rdickson.ucis.dal.ca
Welcome to Glooscap.ace-net.ca - Halifax, Nova Scotia
*****

rdickson@glooscap: ~ $ /usr/local/R/bin/R

R version 2.8.0 (2008-10-20)
Copyright (C) 2008 The R Foundation for Statistical Computing
ISBN 3-900051-07-0

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

    Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> █
```


How Do I get Connected?



- Logging in
 - ssh *Secure Shell*
- Moving Data Around
 - sftp *Secure File Transfer Protocol*
 - scp *Secure Copy*
- both provided with **Mac OS X** or **Linux**
 - ssh -X username@brasdor.ace-net.ca
 - sftp username@placentia.ace-net.ca
- **Windows?** Use PuTTY and WinSCP
just google for them



Getting an account: <http://www.ace-net.ca/>

A screenshot of a web browser window displaying the ACEnet website. The browser's address bar shows the URL "http://www.ace-net.ca/wiki/Get_an_Account". The website's header includes the ACEnet logo and a search bar. A left-hand navigation menu lists various links, with "Get an Account" highlighted by a red circle. The main content area is titled "Get an Account" and contains sections for "Eligibility", "How to Get an Account", and "Step 1. Get a Compute Canada ID". The "Eligibility" section explains that ACEnet facilities are for Canadian researchers and provides a link for non-Canadian researchers. The "How to Get an Account" section instructs users to read the "Conditions of Use" and provides a link to the "Compute Canada Database". The "Step 1" section details the requirement to register with Compute Canada and obtain a CCI. The footer of the page states, "This page was last modified on April 4, 2011, at 14:26."

Firefox

Get an Account - ACEnet

http://www.ace-net.ca/wiki/Get_an_Account

Search

ACEnet

Cluster Status
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Get an Account

Eligibility

ACEnet facilities are designated for Canadian researchers or those collaborating on Canadian research projects. Any academic researcher from a Canadian research institution who needs high performance computing to support his or her research may apply for an account on ACEnet.

Note: If you are a researcher with an organization other than a Canadian University, click [here](#) .

If you are at a Canadian university, continue with the procedure below.

How to Get an Account

Please read [Conditions of Use](#) before you decide to apply for an account.

Step 1. Get a Compute Canada ID

All eligible ACEnet users must [register with Compute Canada](#) , obtaining a Compute Canada Identifier (CCI). Faculty members must obtain a CCI before any non-faculty members of their research group (graduate students, post docs, research assistants, etc.) can register.

Step 2. Get an ACEnet Account

Once you have your CCI (usually takes 24-48 hours) log into the [Compute Canada Database](#)  (CCDB) and click the 'Apply' button next to ACEnet.

This page was last modified on April 4, 2011, at 14:26.

Where to go for help

- ACEnet User Guide
 - http://www.ace-net.ca/wiki/User_Guide
- ACEnet User Wiki
 - <http://www.ace-net.ca/wiki/ACEnet>
- Research/user support personnel
 - email: support@ace-net.ca

What can we do
for you?

What cluster should I use?



- Doesn't matter,
ONE ACCOUNT IS GOOD EVERYWHERE
- Multiple serial jobs? Brasdor best
- Shared memory parallel? Fundy best
- MPI? Mahone best, Placentia, Glooscap, Fundy good
- Some software has only local license
 - e.g. Matlab DCS, Gaussian only at Placentia
 - See <http://wiki.ace-net.ca> for availability, or
contact your Computational Research Consultant

Organization and funding

- Consortium of Atlantic Cdn. universities
- Compute Canada is nat'l. umbrella org.
- Several \$million from
CFI + ACOA + provinces + “in kind”
- 10 technical staff @ 5 sites
- Research Directorate
Local User Groups ← user direction!
Resource Allocation Committee

Member institutions

